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Deep hybrid model for Mpox disease diagnosis from skin lesion images

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Abstract

This research presents DNLR-NET, a novel model designed for automated and accurate diagnosis of MPox disease. The model's performance is constructed and validated using a carefully collected MPox dataset from online repositories. DNLR-NET begins by extracting deep features from the DenseNet201 pre-trained model, which exhibited superior performance compared to other models during the comparison. The deep features obtained from each dense layer are then used to train six classifiers, among which logistic regression showcases the best performance with the extracted deep, dense feature. A comparative study with earlier advanced CNN models classifying the same dataset demonstrates that DNLR-NET achieves an impressive accuracy of 97.55%, outperforming the base DenseNet201 model, which only attains 95.91% accuracy. This accuracy emphasizes the efficacy of combining deep features with logistic regression. A Grid Search algorithm is employed for optimal hyperparameter extraction, creating multiple unified deep feature sets and achieving the highest classification accuracy. The fusion of deep features with logistic regression yields superior results compared to ensemble techniques such as random forest and support vector machines and also reduces training time complexity. DNLR-NET surpasses existing models, ML classifiers, and pre-trained models, demonstrating its effectiveness and potential for clinical implementation in diagnosing MPox. The promising outcomes of advantage deep learning algorithms, particularly DenseNet201 transfer learning, highlight the significance of adopting transfer learning methodologies for CNN-based MPox diagnosis in clinical settings. Researchers and clinicians are strongly encouraged to explore and implement these techniques to improve the accuracy and efficiency of MPox diagnosis.

KEYWORDS

DenseNet201, logistic regression, machine learning, MPox, multiclass

1 | INTRODUCTION

Over the past two decades, the world has faced several significant viral disease outbreaks, including the devastating COVID-19 pandemic that originated in 2019. The global community is still grappling with the aftermath of this novel coronavirus outbreak,[1] with recovery efforts ongoing. However, in 2022, a new threat emerged in the form of MPox, indicating the emergence of yet another worldwide virus.[2] This development has raised concerns

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among health authorities and the general population, as the impact of the previous pandemic is still being felt.

MPox is caused by the Zoonotic Ortho poxvirus, closely related to cowpox and smallpox, and belongs to the Poxvirus family, specifically the genus Ortho poxvirus. While monkeys and rodents primarily transmit the virus, human-to-human spread is also prevalent. The first[3] human case of MPox was identified in the Democratic Republic of the Congo in 1970 during efforts to eradicate smallpox. Historically, MPox has been more commonly observed in central and western Africa, particularly among individuals living near tropical rainforests. In recent times, however, MPox has extended its reach beyond Africa, affecting individuals in urban regions in other parts of the world.[4] The rapid propagation of MPox across 19 countries outside the initial outbreak zones has become a serious global health concern. Given the ongoing impact of the COVID-19 pandemic,[5] the world can only afford to face another disease outbreak with effective measures in place.

Early diagnosis of MPox is vital for curbing its spread, isolating infected cases, and providing timely and appropriate treatment.[6] However, traditional diagnostic methods have limitations, such as the polymerase chain reaction (PCR) technique or electron microscopy for skin lesion checks. PCR[7] can be expensive and time-consuming, and in rural areas, access to sufficient testing kits is often challenging. Additionally, the rareness of MPox and the similarity of skin lesions to other diseases, such as chickenpox and measles, make early recognition through electron microscopy images complex.

To address these challenges, automated approaches[8] driven by artificial intelligence (AI) have emerged as essential tools in disease detection and computer-aided diagnosis (CAD). The global response to the COVID-19 pandemic has highlighted the importance of accurate and timely disease detection to prevent further outbreaks and mitigate morbidity. AI techniques have proven successful in diagnosing various diseases, including breast cancer,[7,9] brain tumors,[10,11] and other diseases. Moreover, AI has shown promising applications in renewable energy, industry, and agriculture. The utilization of Machine Learning (ML) techniques, with a particular emphasis on deep learning (DL), is increasingly prevalent in current research. Within this landscape, Transfer Learning (TL),[12] a specialized subset of ML, plays a crucial role in medical imaging and diagnosis. For instance, previous studies have used TL approaches to detect diseases like malaria, breast cancer, and skin diseases with impressive results.

Although there have been some studies on MPox detection[13] using TL, the existing literature has several limitations. Many of these studies have focused on binary classification,[14–17] which needs to be revised for the multiclass nature of MPox diagnosis. Additionally, model interpretation techniques[18] have often been overlooked, making it challenging to gain insights into the model's decision-making process and identify regions potentially affected by the virus.

This research aims to develop an AI-driven system for diagnosing MPox using a hybrid approach. The proposed model will utilize a range of pre-trained DL models with traditional ML classifiers and optimizer combinations to achieve optimal performance. Moreover, model interpretability will be a focal point, ensuring that health practitioners can trust and rely on the system for mass screening and diagnosis.

The following is the paper's main contributions:

1. The paper introduces DNLR-NET, a novel model designed explicitly for accurate automated diagnosis of MPox skin infection. The DenseNet201 pre-trained model extracted deep features from the MPox dataset. These features have been integrated into the ML classifier, which significantly achieving high accuracy.
2. A Grid Search algorithm is used to identify the optimal hyperparameters, creating unified deep feature sets. This optimization process significantly enhances the classification accuracy of the DNLR-NET model.
3. Among the six classifiers explored, logistic regression emerges as the best-performing one using the extracted deep features. This choice of classifier is highlighted as a critical factor contributing to the success of the DNLR-NET model.
4. The DNLR-NET model achieves an impressive accuracy of 97.60%, significantly outperforming the base DenseNet201 model (95.91% accuracy). This notable improvement demonstrates the efficacy of the DNLR-NET approach for MPox diagnosis.

In the subsequent sections of this paper, we will present the literature research methodology, describe the dataset used, outline the experimental setup, and present the obtained results. Furthermore, we will discuss and draw conclusions based on the findings. By overcoming the limitations of previous studies and providing a comprehensive AI-driven solution for MPox diagnosis, this research aims to contribute significantly to global efforts in combatting infectious diseases and ensuring public health safety.

2 | RELATED WORK

The detection and diagnosis of human MPox[19–21] have received increasing attention in the scientific literature,

particularly with the recent global spread of MPox infection. However, research is insufficient in applying CAD[22] for primary diagnosis due to the limited availability of publicly accessible datasets for training and testing purposes. CNNs[23] have demonstrated remarkable ability in analyzing medical images and identifying patterns related to various diseases, making them promising candidates for MPox diagnosis. To address the lack of data and enhance the efficiency of DL models, researchers have widely adopted TL, which involves repurposing pre-trained models trained on large datasets for similar tasks, leading to improved model performance.

Previous studies[13] utilized TL to develop a modified version of the VGG CNN model. In contrast, others explored various pre-trained CNN architectures, such as ResNet-18, MobileNet, NasNetMobile, GoogLeNet, EfficientB0, and ShuffleNet. These studies achieved promising results on the MPox Skin Lesion Dataset (MSLD)[24] and the MPox Skin Image Dataset (MSID).[25] Furthermore, novel approaches integrating AI-based optimization techniques have been proposed for MPox diagnosis. These approaches involve fine-tuning customized CNN layers using AI-Biruni Earth Radius (BER)[3] optimization-based stochastic fractal search (BERSFS)[22] and employing optimization algorithms like sine cosine (SC) and particle swarm (PS) for feature selection with multilayer perceptron (MLP) classifiers. Additionally, several studies utilized various CNN architectures[13] to differentiate between MPox and non-MPox cases using both MSID and MSLD datasets. Customized CNN models were developed by gathering data from web searches of available skin images. Generalization and Regularization (GR)-based TL[24] were employed to distinguish between MPox and normal images.

In the domain of DL-based classification of MPox disease, several studies have focused on classifying MPox skin lesions using DL methods, such as VGG-16,[26] ResNet50,[27] and InceptionV3.[28] Another study utilized the Xception[29] TL model in combination with Grad-Cam and LIME techniques, achieving high accuracy and F1-score. TL was also employed to detect MPox disease using a modified VGG16 model. Researchers employed TL methods on image data, achieving impressive accuracy using the MobileNetv2[24] model in the context of human-monkey disease classification from skin lesion images via a mobile application. Additionally, a study utilized the residual architecture[27] to diagnose Lyme disease from skin lesion images, demonstrating the potential of combining CNN and TL in skin lesion image classification.

These studies illustrate the promising potential of DL and TL methods for medical image classification, including detecting infectious diseases like MPox[3] and Lyme disease.[27] However, challenges related to model interpretability, generalization on different datasets, and computational efficiency remain, requiring further research. Furthermore, this paper presents a comprehensive investigation into applying AI-driven DL models for precise and reliable MPox detection by a hybrid learning approach, aiming to develop an interpretable and efficient framework capable of multiclass classification. By providing a valuable tool for timely diagnosis and containment of MPox, this research aims to contribute to the global effort to mitigate its impact on human health.

3 | MATERIALS AND METHODS

This section provides a comprehensive overview of the methods and materials used in our study to classify multiclass MPox skin infections. The primary approach employed for this task is based on the DNLR-NET architecture. We elaborate on the different components of our methodology in the following subsections.

3.1 | MPox datasets

Our proposed approach for developing and evaluating the DNLR-NET model relied on a comprehensive MPox imaging dataset. The dataset was carefully curated to encompass diverse skin infection cases, ensuring representation from various disease classes. It consists of high-resolution images, each accompanied by corresponding class labels denoting the specific disease category.

During the data acquisition process, the author[13] meticulously selected images from various sources, including online research centers, specialized websites, and repositories focused on infectious diseases and dermatology. This approach compiles a wide range of cases, capturing the heterogeneity of MPox manifestations across different geographical regions. Figure 1 illustrates a sample from each class and the total number of images for each class.

3.2 | Preprocessing and augmentation

We subjected the dataset to preprocessing steps to ensure data quality and facilitate efficient model training and evaluation. This included standardization of image resolution, removal of artifacts, and normalization of pixel values. The subjected dataset has distortion near the infection edge due to acquisition from different online resources. These preprocessing steps were essential to achieve uniform characteristics across the dataset, enabling seamless integration into the DNLR-NET

FIGURE 1 MPox dataset.

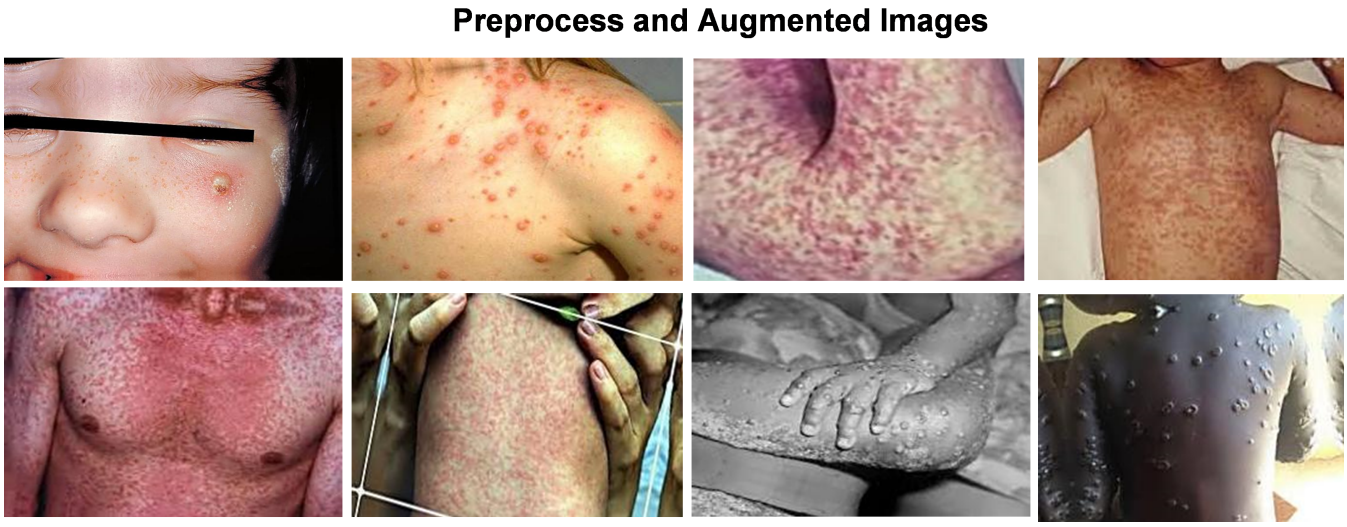

FIGURE 2 Preprocess & augmented samples.

TABLE 1 Dataset distribution.

Class	Train	Test	Total
Chickenpox	677	169	846
Measles	584	146	730
Monkeypox	602	150	752
Normal	596	149	745

model. Furthermore, the dataset was meticulously labeled with class annotations to correspond to the specific skin lesion categories. These annotations served as ground truth information for training and validating the DNLR-NET model, enabling the model to classify MPox skin infections accurately.

Additionally, image augmentation techniques such as rotating image (90°), Flip (Vertical or horizontal), and Zoom (In or Out) have been applied to generalize DNLR-NET learning. Figure 2 displays sample images after preprocessing and augmentation. The dataset was partitioned into training and testing sets to enable effective model training and unbiased evaluation. A significant portion of the data was used for training the DNLR-NET model. At the same time, separate sets were reserved for validation and testing to assess the model's generalization ability. Table 1 presents the dataset distribution after augmentation.

As a result, the dataset size for the Mpox, chickenpox, measles, and normal classes has been expanded, with 752, 846, 730, and 745 images, respectively. The augmented dataset now comprises a total of 3073 images, which have been split into separate training (80) and testing sets (20). The training set consists of 2459 images, while the testing set contains 614 images. Each image is appropriately labeled with its corresponding class. Table 1 shows the sample distribution for both the training and testing datasets.

3.3 | Methodology

In this study, we adopt a two-step approach for classifying MPox skin infections, advantage the strengths of pre-trained models and ML classifiers. Firstly, we explore using the DenseNet201 pre-trained model as a feature extractor. DenseNet201 has demonstrated superior performance in image recognition tasks, making it a promising choice for feature extraction in our classification task. We aim to capitalize on its ability to capture intricate and

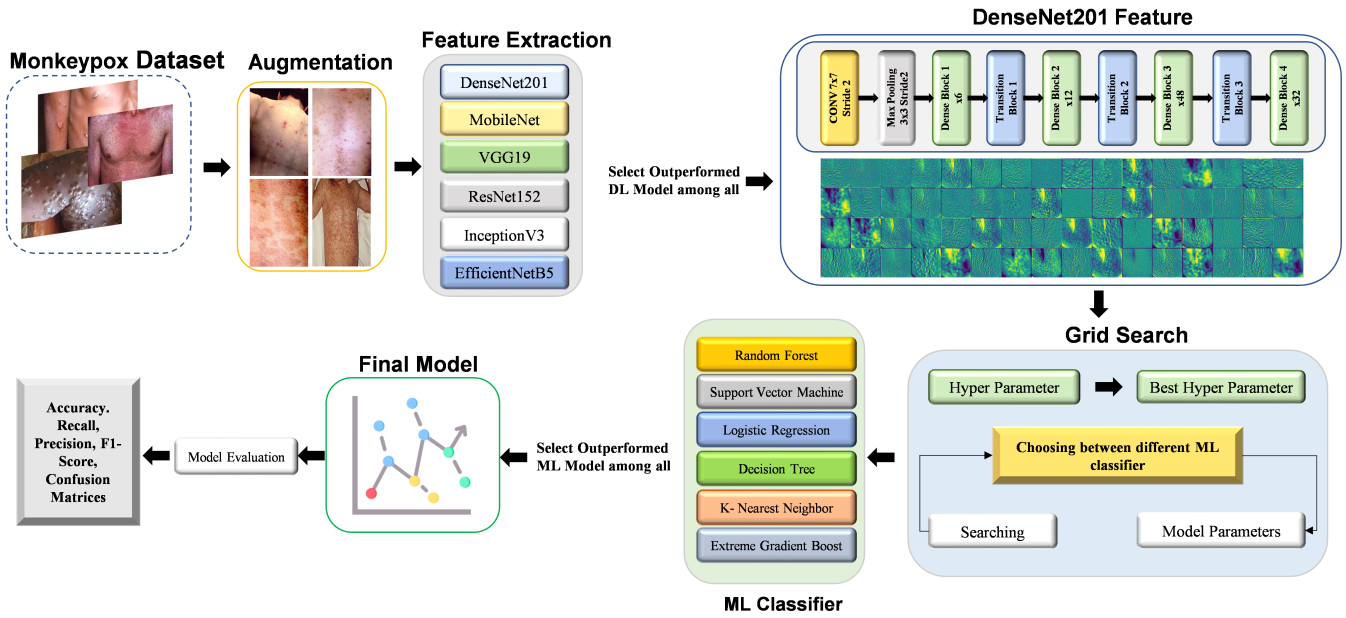

FIGURE 3 Proposed DNLR architecture diagram.

relevant features from input images, which can significantly enhance the accuracy of our classification DNLR-NET model.

In the second step of our approach, we employ ML classifier to utilize the extracted features for final class prediction. Before applying the ML classifier, we employ the Grid Search technique to optimize hyperparameters. Grid Search enables us to exhaustively search through a range of hyperparameter values, systematically evaluating their impact on the classifier's performance. By tuning the hyperparameters through Grid Search, we can identify the most suitable configuration, thus enhancing the classifier's ability to generalize and achieve higher accuracy in classifying MPox skin infections.

By integrating the feature extraction capabilities of DenseNet201 and the hyperparameter optimization using Grid Search with the ML classifier, we aim to develop a robust and precise classification model. This two-step approach allows us to maximize the potential of both pre-trained DL models and fine-tuned ML classifiers, yielding a robust and reliable method for accurate MPox skin infection classification. In the subsequent sections, we provide comprehensive details of our experimental setup, implementation of DenseNet201 as a feature extractor, the application of Grid Search for hyperparameter tuning, ML classifier selection, and evaluation metrics utilized to assess the performance of our proposed methodology as shown in Figure 3.

3.3.1 | Pre-trained models

Using pre-trained models for skin infection prediction via image datasets offers significant advantages in medical image analysis. These models, pre-trained on large-scale image datasets from diverse sources, have already learned to extract meaningful features from images, including intricate patterns and structures specific to skin infection. By advantage this knowledge, we can effectively classify MPox infection with improved accuracy and reduced training time. Moreover, pre-trained models demonstrate excellent generalization capabilities, enabling them to perform well on unseen infection images and even different types of skin infection. This data-efficient approach is precious in medical applications, where acquiring large labeled datasets can be challenging. Additionally, the availability of pre-trained models as open-source resources fosters collaboration and knowledge sharing among researchers and medical practitioners, further advancing diagnosis and treatment.

In skin infection prediction, pre-trained models serve as a foundation for transfer learning, allowing us to fine-tune the models on our specific dataset. This process adapts the models to recognize skin infection-specific features, leading to improved performance on the prediction task. Furthermore, pre-trained models demonstrate superior feature extraction capabilities, crucial for accurately identifying infected skin from medical images. Their ability to learn rich and complex representations from diverse image data contributes to achieving high accuracy in classification, even when faced with limited training data. By harnessing the benefits of pre-trained models, researchers, and medical professionals can enhance the efficiency and accuracy of MPox prediction, ultimately leading to more effective diagnosis and treatment strategies.

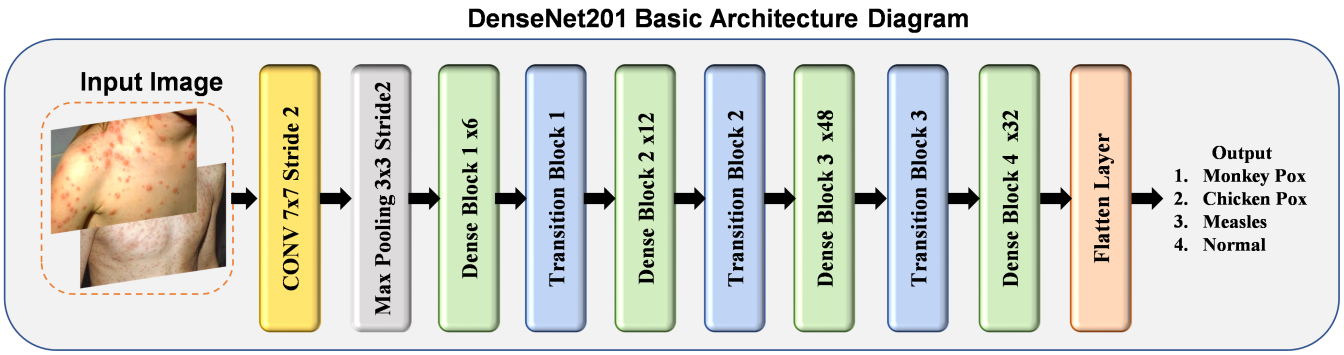

FIGURE 4 DenseNet201 basic architecture diagram.

DenseNet201

DenseNet201 offers several technical advantages over other pre-trained models for image classification tasks. The key differentiating feature of DenseNet201 is its densely connected architecture, where each layer is directly connected to all subsequent layers within a dense block. This dense connectivity promotes feature reuse and information flow, allowing the model to capture intricate patterns and dependencies in the data effectively. As a result, DenseNet201 requires fewer parameters than other deep models like VGG or ResNet, making it more memory-efficient and computationally faster during training and inference. The dense skip connections also mitigate the vanishing gradient problem, enabling smoother and more efficient training convergence. DenseNet201's ability to exploit feature reuse and information flow and its reduced computational complexity make it a highly effective and practical choice for skin infection prediction and other image classification tasks, particularly in scenarios with limited resources and large-scale datasets. Figure 4 illustrates the basic architecture of the DenseNet201 model.

3.3.2 | Machine learning classifier

Integrating an ML classifier with pre-trained model features offers several advantages over using other pre-trained models directly. While pre-trained models are adept at feature extraction from images, they may need to be specifically optimized for the target task, such as MPox prediction. Using an ML classifier with pre-trained model features, we can fine-tune the feature representations to suit the skin infection classification task, enhancing the model's performance. This fine-tuning process allows the ML classifier to learn task-specific patterns and relationships in the extracted features, improving the accuracy and generalization of MPox images. Additionally, using an ML classifier provides flexibility in model selection, allowing us to experiment with various classifiers and identify the one that best suits the dataset and classification requirements. The pre-trained model's robust feature extraction capabilities with an ML classifier's adaptability and optimization potential result in a powerful approach for accurate and efficient MPox prediction.

Logistic regression

The benefit of using a logistic regression classifier Equation (1) with pre-trained model features lies in its simplicity, interpretability, and efficiency compared to other pre-trained models. While complex DL architectures like DenseNet or ResNet may offer superior performance in specific tasks, they often require substantial computational resources and longer training times. In contrast, logistic regression is a linear classifier that works well with extracted features from pre-trained models, making it computationally lightweight and faster to train.

Moreover, logistic regression provides interpretable results by estimating the probability of MPox presence based on the extracted features. This interpretability is crucial in medical applications, as it allows clinicians to understand the contribution of each feature to the final prediction, aiding in diagnosis and decision-making. Additionally, logistic regression is less prone to overfitting, making it suitable for datasets with limited samples, which is common in medical imaging datasets. The combination of logistic regression with pre-trained model features strikes a balance between performance, interpretability, and efficiency, making it a valuable choice for MPox prediction tasks in medical image analysis.

In Equation (1), π represents the "success probability," indicating the probability of an observation belonging to a specific category of a binary variable. The model captures the likelihood of an event occurring based on various

TABLE 2 Optimal hyper parameter of ML classifier search by grid search.

ML classifier	Optimal hyper parameter
Random forest	Random_state:30, max_features: auto, n_estimators:500, max_depth:8, criterion: gini
K nearest neighbor	leaf_size:1, n_neighbors:8, weights: distance
Support vector machine	gamma: 0.0001, kernel: linear
Logistic regression	C:1.0, penalty: l1

factors, such as x. For example, it can estimate the probability that an individual is afflicted with the Mpox virus. Notably, estimates of π obtained from equations like the above always fall within the range of 0 to 1 when employing the logistic model. This occurs because $e^{(\beta_0+\beta_1 x)}$ is a positive exponent of the base e, necessitating a positive numerator. Furthermore, since the model's denominator is the sum of its numerator and denominator, the result is consistently less than 1.

$$\pi(x) = \frac{e^{(\beta_0+\beta_1 x)}}{1+e^{(\beta_0+\beta_1 x)}} \tag{1}$$

Optimal hyper parameter search via grid search

Hyperparameter optimization is a conventional technique to fine-tune ML algorithms for optimal performance. One common method, Grid Search, systematically explores a predetermined range of hyperparameter values. Appropriate boundaries are often set for specific parameters in the ML approach to ensure efficiency and prevent infinite or invalid parameter values. Though Grid Search can be complex in high-dimensional spaces, it offers parallelizability. Table 2 presents the best hyperparameter configurations obtained through the Grid Search process for the ML classifiers.

Furthermore, we ensured consistency by employing the best hyperparameters for training the backbone DenseNet201 based TL model. Specifically, we conducted training with 10 epochs, a learning rate of 0.0001, and a batch size 64. Exploiting the efficiency of the Adam optimizer, we observed notable improvements in model performance. The Adam optimizer exhibited superior accuracy and minimized memory usage compared to alternative optimizers. To evaluate and enhance overall performance effectively, we adopted the categorical cross-entropy loss function while training the DenseNet201 model, emphasizing its crucial role in our approach.

4 | EXPERIMENTS AND RESULTS

This section presents the experimental setup and results obtained from the DenseNet201 pre-trained model as the primary feature extracted with logistic regression as the final prediction. Additionally, we conduct a comparative analysis of the TL techniques, ML classifier, and the proposed model's effectiveness with the existing state-of-art model. For evaluation purposes, we utilize the widely-used MPox[24] dataset, which is publicly available and has been commonly employed as a benchmark in previous studies for performance assessment. Our primary objective is to enhance the accuracy of the multiclass skin lesion recognition process through our proposed model. To validate our proposed model efficacy, we have used benchmarked performance metrics: Accuracy (Equation 2), Precision (Equation 3), Recall (Equation 4), and F1-Score (Equation 5).

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{2}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{3}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{4}$$

$$F1-\text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision}+\text{Recall}} \tag{5}$$

4.1 | Experimental setup

The DNLR-NET model in this research was trained using the Python programming language and the Keras framework. All experiments were conducted in the widely-used Google Colab notebook environment with the GPU runtime type to take advantage of its computational capabilities. Google Colab provides free access to researchers, making it a convenient platform for running tests. The model training and testing were performed on an NVIDIA GeForce MX350 Tesla GPU with 8 MB RAM.

4.2 | Performance analysis of proposed DNLR-NET architecture

This section presents the performance analysis of our proposed DNLR-NET model. We will discuss the classification performance metrics of the DNLR-NET model, which utilizes deep features obtained from the DenseNet201

Models	Precision	Recall	F1-Score	Accuracy
DenseNet201 (Base)	96.25	95.91	95.92	95.91
Logistic Regression + DenseNet201	**97.63**	**97.60**	**97.60**	**97.55**

TABLE 3 Classification performance of proposed DNLR model.

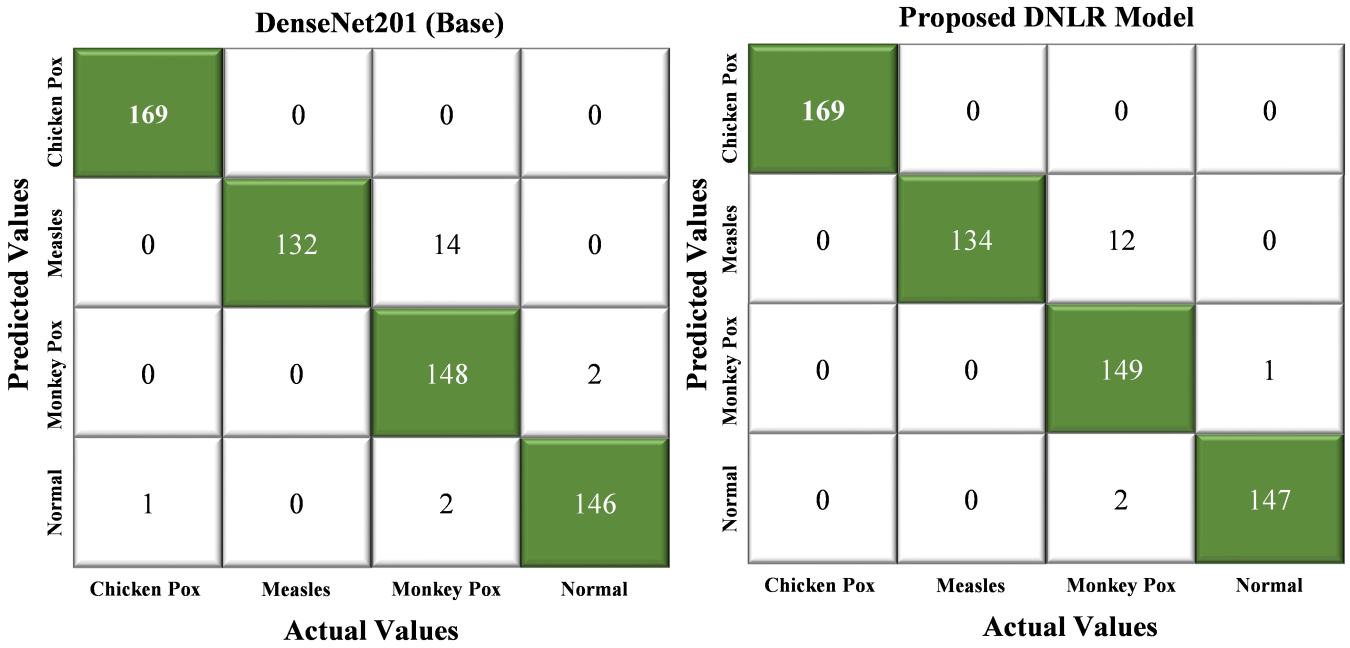

FIGURE 5 Actual Vs predicted accuracy via confusion matric for DenseNet201 (Base), and DNLR Model.

pre-trained model used in this study. Also, confusion matrices will show diagnostic accuracy, comparing actual vs. predicted results. Furthermore, we will compare our proposed model with eight other pre-trained models and existing CNNs. Lastly, we will analyze the feature map to evaluate the model interpretability as a deep feature extractor.

4.2.1 | Classification performance

Table 3 presents the performance metrics of two models: DenseNet201 (Base) and Logistic Regression + DenseNet201. These models were evaluated using precision, recall, F1-score, and accuracy to assess their classification performance. DenseNet201 (Base), the model achieved a precision of 96.25, recall of 95.91, F1-score of 95.92, and an overall accuracy of 95.91. While Logistic Regression + DenseNet201 (DNLR-NET), the model demonstrated improved performance with precision, recall, and F1-score, 97.63, 97.60, and 97.60, respectively, with a significantly higher accuracy of 97.60%. Train weight of backbone DenseNet201 incorporated with Logistic Regression to enhance classification capabilities, as evidenced by the higher precision and recall values. The F1-score and recall of DNLR remained consistent at 97.55, indicating a balanced performance. The substantial increase in accuracy demonstrates the effectiveness of the combined model in making correct predictions.

Furthermore, confusion matrices are utilized to evaluate the performance of the DNLR-NET model. Figure 5 displays the confusion matrices, and a comparison is made between the DenseNet201 (Base) as a feature-extracted model and the DNLR-NET model in terms of their prediction accuracy for different infection class categories. The results indicate that the DenseNet201 (Base) model misclassified 19 samples, with 14 misclassifications of the Measles class and two to the MPox class, and three misclassified of normal. In contrast, the DNLR-NET model achieved significantly better performance, with deep features with only 15 misclassifications out of 614 samples, demonstrating its improved accuracy and effectiveness in classifying infection types. The significant reduction in misclassifications observed in the DNLR-NET model compared to the DenseNet201 (Base) model indicates the superiority of the DNLR-NET model's classification capabilities, making it a promising approach for infection classification tasks.

4.2.2 | Performance comparison of proposed DNLR-NET architecture with existing methods

Table 4 shows the accuracy of different convolutional neural network models. The proposed model, DNLR-NET, achieves the highest accuracy of 97.60%. DNLR-NET is a deep neural network with logistic regression that uses a novel architecture to achieve high accuracy. The model is based on the DNLR-NET architecture, which combines DenseNet201 features with a logistic regression classifier. The other models in Table 4 also achieve high accuracy, with MobileNet achieving 91.11%, Inception+Xception+DenseNet-169 + Beta function-based normalization method achieving 93.39%, EfficientB0 + DenseNet achieving 94.57%, and GoogLeNet+BER + SC + PS+ MLP achieving 93.8%. These models use various techniques to achieve high accuracy, including depth-wise separable convolution, beta function-based normalization, lightweight architectures, and Bayesian error reduction. Table 4 shows that the proposed DNLR-NET is an effective tool for skin infection classification with the highest accuracy.

TABLE 4 Performance comparison with exiting advance CNN's model.

Reference	Model	Accuracy
Sahin et al.[14]	MobileNet	91.11%
Pramanik et al.[28]	Inception+Xception +DenseNet-169 + Beta function-based normalization method	93.39%
Liu et al.[30]	EfficientB0 + DenseNet	94.57%
Abdelhamid et al.[22]	GoogLeNet+BER + SC + PS+ MLP	93.80%
Proposed model	DNLR	97.55%

4.2.3 | Performance comparison of proposed DNLR-NET architecture with pre-trained models

Table 5 compares various pre-trained DL models and the proposed DNLR-NET model based on their performance metrics. The models are evaluated in precision, recall, F1-score, and accuracy. Precision represents the proportion of true positive predictions among all positive predictions, indicating the model's ability to avoid false positives. The DNLR-NET model achieves the highest precision score of 97.63, indicating that it can make correct positive predictions. Recall, also known as sensitivity or true positive rate, signifies the proportion of true positive predictions out of all actual positive instances. Again, the DNLR-NET model stands out with a recall score of 97.60, demonstrating its proficiency in capturing the most positive instances. F1-score is the harmonic mean of precision and recall, providing a balanced measure of the model's performance. The DNLR-NET model excels in F1-score with an impressive score of 97.60, indicating a well-balanced trade-off between precision and recall. Lastly, accuracy represents the overall correctness of the model's predictions. The DNLR-NET model attains the highest accuracy of 97.60, outperforming all other pre-trained models ranging from 89.28 to 95.74. Overall, the DNLR-NET model showcases superior performance across all metrics, making it a promising and highly accurate solution for the given task when compared to the other base models like DenseNet-169, EfficientNet-B0, EfficientNet-B5, EfficientNet-B7, NASNet-Mobile, Xception, and Inception-V3.

4.2.4 | Performance comparison of proposed DNLR-NET architecture with ML classifier

Table 6 presents the evaluation results of different models in conjunction with the DenseNet201 pre-trained model.

TABLE 5 Performance comparison with pre-trained models.

Models	Precision	Recall	F1-Score	Accuracy
DenseNet-169 (Base)	94.29	94.21	94.15	94.21
EfficientNet-B0 (Base)	96.28	95.74	95.81	95.74
EfficientNet-B5 (Base)	95.03	94.89	94.91	94.89
EfficientNet-B7 (Base)	94.93	94.89	94.89	94.89
NASNetMobile (Base)	89.90	89.28	89.32	89.28
Xception (Base)	95.23	95.23	95.24	95.23
Inception-V3 (Base)	95.27	95.23	95.23	95.23
DNLR model	97.63	97.60	97.60	97.55

Models	Precision	Recall	F1-Score	Accuracy
RF + DenseNet201	90.06	87.12	87.12	87.13
Gaussian NB + DenseNet201	82.10	68.09	68.09	68.09
SVM + DenseNet201	94.12	92.10	92.10	92.10
DNLR (LR + DenseNet201)	97.63	97.60	97.60	97.55

TABLE 6 Performance comparison with ML classifier along dense feature.

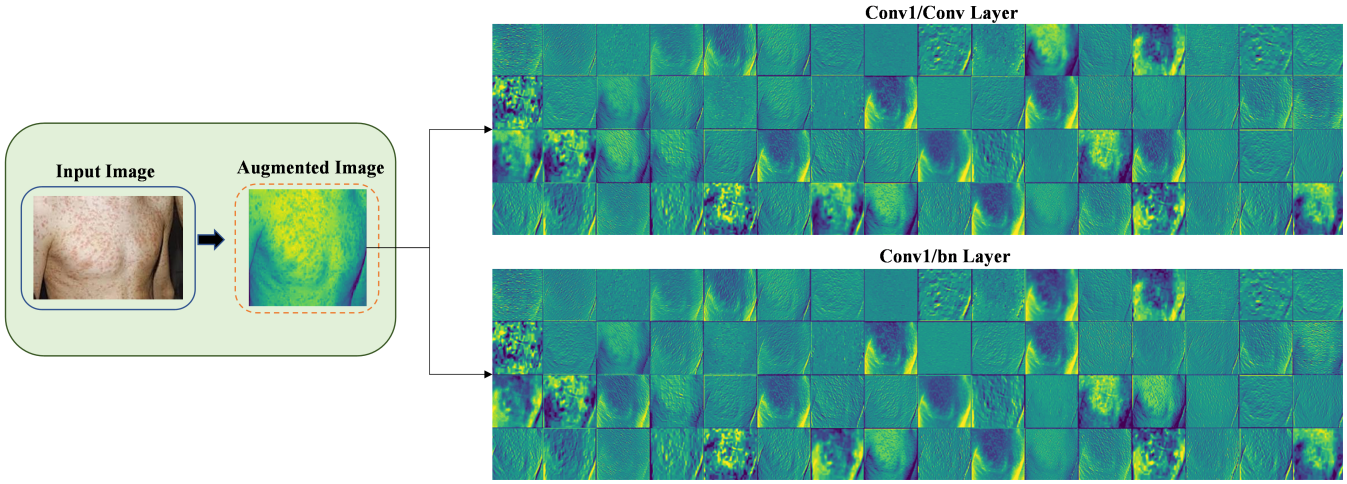

FIGURE 6 Feature map at different convolution layer in DenseNet201 (Base) Model.

Each row corresponds to a specific model, and the columns represent various performance metrics, including Precision, Recall, F1-Score, and Accuracy. The RF + DenseNet201 model achieved a Precision of 90.06, Recall of 87.12, F1-Score of 87.12, and an overall Accuracy of 87.13. The Gaussian NB + DenseNet201 model showed a Precision of 82.10, a Recall of 68.09, an F1-Score of 68.09, and an Accuracy of 68.09. The SVM + DenseNet201 model performed well, achieving a Precision of 94.12, Recall of 92.10, F1-Score of 92.10, and Accuracy of 92.10.

However, the DNLR-NET Model stands out with exceptional performance, attaining a Precision of 97.63, Recall of 97.60, F1-Score of 97.60, and an impressive Accuracy of 97.55. This result demonstrates that the DNLR-NET Model outperforms the other models significantly, making it a superior choice for the given task. It shows the potential of the proposed DNLR-NET Model to effectively manipulated the deep features obtained from the DenseNet201 (Base) pre-trained model, resulting in a highly accurate and robust classification system.

The rationale behind model selection becomes evident when analyzing the results presented in Tables 3, 5, and 6, which compare several pre-trained base models and a hybrid model, each integrating DenseNet201 as the base with various ML classifiers, in terms of their accuracy scores. Among the base models, DenseNet201 achieved the highest accuracy at 95.91%, closely followed by EfficientNetB0 with an accuracy of 95.74%. However, the hybrid model significantly outperformed all base models, demonstrating an exceptional accuracy of 97.60%. Interestingly, when integrating DenseNet201 predicted weights with different ML classifiers, only logistic regression managed to outperform the others. The fusion of DenseNet201's feature representation capabilities and feature propagation with logistic regression, a simple yet interpretable classifier, creates an efficient synergy that manipulates the strengths of both techniques. DenseNet201, renowned for its proficiency in extracting high-level features from complex data, harmonizes seamlessly with logistic regression. This combination enhances predictive accuracy while providing valuable insights into the significance of individual features, rendering it particularly well-suited for diagnosing MPox.

4.3 | Feature map analysis

Feature map analysis is a visualization technique in computer vision and DL, providing valuable insights into the inner workings of CNNs. At its core, Figure 6 feature map analysis involves examining and interpreting the activation patterns generated by different filters or neurons within a CNN layer of DenseNet201 due to connections

between each layer, effectively addressing the vanishing-gradient problem and enhancing feature propagation. This architecture excels at representing complex features, making it well-suited for DL tasks. Each filter is designed to detect specific features or patterns in an input image, and the resulting feature maps represent the responses of these filters to various aspects of the input. By visualizing and analyzing these feature maps, researchers and practitioners gain a deeper understanding of what features or patterns each filter is capturing. This process is instrumental in diagnosing DenseNet201 model behavior prediction integration with other ML classifiers. Figure 6 depicts the visualization of intermediate feature maps from various convolution blocks for the base model of DenseNet201. This visualization enables us to observe the decomposition of inputs into different filters learned by the model. The results demonstrate that the DNLR-NET model's basic architecture captures dense features from the input image. This study employs feature map analysis techniques to assess the model interpretability. Using the DenseNet201 model, it does feature importance analysis to determine which features are most important during training and how they affect model predictions. Additionally, Figure 6 visualizes feature interactions. Moreover, the model's predictions are explained in human-understandable terms, enabling users to comprehend the rationale behind the model's decisions.

4.4 | Discussion and future work

The investigation of MPox illness encountered many constraints that extended beyond the scope of this study. A primary challenge arose from the limited availability of data, as there was no access to an extensive dataset about MPox. This scarcity of data posed a significant impediment to data authentication, interpretation, and utilization, thereby hindering the establishment of a robust benchmark literature for the study. Furthermore, the inadequacy of pertinent literature could have helped our ability to conduct a comprehensive assessment and accurately evaluate the performance of the proposed DNLR-NET model. A more diverse and expansive dataset would have facilitated a more comprehensive evaluation of the model's capabilities and effectiveness. A primary factor contributing to the limited data accessibility was privacy concerns. Given that MPox often affects the entire body, obtaining and utilizing photographs of individuals afflicted by the disease, particularly children, proved to be a challenging endeavor. The imperative need to safeguard the privacy and sensitivity of the subjects in medical imaging posed substantial obstacles in procuring a substantial dataset for analysis.

Despite these formidable challenges, concerted efforts are underway to mitigate the issue of data scarcity. Various public repositories continuously collect new images, promising to bolster the available information for research purposes. This ongoing data accumulation has the potential to ameliorate data availability in the future, affording more robust investigations into the efficacy of the proposed DNLR-NET model. Additionally, another significant area warranting attention in the study of MPox is its rarity, which necessitates increased expertise among healthcare professionals. The dearth of specialists in this field can result in misdiagnoses, delayed treatments, and the potential for disease outbreaks, escalating into epidemics.

The dataset employed in this study primarily emanated from public sources rather than medical centers or hospitals. However, this dataset consisted of only a relatively limited number of images from a small cohort of distinct patients. Consequently, the constrained size and lack of diversity in the dataset curtail the applicability of the DNLR-NET model's findings to a broader demographic.

Furthermore, it is imperative to acknowledge that the study of MPox illness grapples with substantial complexities arising from data scarcity, a need for more relevant literature, privacy constraints, and the rarity of the disease. Notwithstanding these limitations, ongoing endeavors to amass more extensive datasets hold the potential to catalyze more comprehensive research and enhance the assessment of proposed models such as DNLR-NET. Addressing the need for heightened expertise among healthcare professionals in this domain is crucial to combat and manage the disease effectively.

In future research, we intend to address this concern by augmenting our datasets with additional samples, incorporating a broader spectrum of patient images from diverse healthcare facilities and hospitals. This approach is poised to enhance the generalizability of our model and contribute to a more robust understanding of MPox illness.

5 | CONCLUSION

This study introduced a novel DNLR-NET model for the automated and accurate diagnosis of MPox disease. The performance of the DNLR-NET model was constructed and validated using an MPox dataset collected from online sources. The DNLR-NET model begins by extracting deep features from the DenseNet201 pretrained model, which outperformed other models in the comparison. These deep features are obtained from each dense layer and are utilized to train six classifiers. Only logistic regression with the extracted deep dense feature

demonstrated superior performance among these classifiers. A comparison study was conducted to evaluate the performance of DNLR-NET against earlier advanced CNN models that classified the same dataset. The results showed that DNLR-NET achieved an accuracy 97.55, while the base DenseNet201 model achieved 95.91 accuracy. This accuracy highlights the effectiveness of combining deep features with logistic regression. A Grid Search algorithm optimized feature extraction to create multiple unified deep feature sets. This approach yielded the best available hyperparameters and contributed to the highest classification accuracy. Combining deep features with logistic regression proved superior to ensemble techniques like random forest and support vector machines. Logistic regression reduced the training time complexity, making it a favorable choice for the DNLR-NET classification model.

The proposed DNLR-NET method demonstrated superior performance in diagnosing MPox compared to prior advanced CNN models. It outperformed most existing CADs, ML classifiers, and pre-trained models, establishing its effectiveness and potential for clinical implementation. The promising results of the DNLR-NET model, advantage DL algorithms such as DenseNet201 transfer learning, encourage researchers and clinicians to adopt TL methodologies for CNN-based MPox diagnosis in clinical settings.

AUTHOR CONTRIBUTIONS

Saif Ur Rehman Khan & Sohaib Asif: Conceptualization, methodology, data curation, writing-original draft preparation, investigation, validation, visualization, writing. Omair Bilal & Sajid Ali: Reviewing and editing.

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CONFLICT OF INTEREST STATEMENT

There is no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the authors upon reasonable request and with permission of the Author.

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